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United States Patent	12397844
Kind Code	B2
Date of Patent	August 26, 2025
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Input device

Abstract

An input device including a rotatable input unit by which it is possible to perform a plurality of types of operations is provided. The input device includes an input unit that can rotate about a rotational center shaft; an electrostatic sensor provided around the input unit; and an operation position detecting unit configured to detect, based on an output of the electrostatic sensor, a position of an operational input that is performed on the input unit or around the input unit.

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Appl. No.: 18/468034

Filed: September 15, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240001982 A1	Jan. 04, 2024

Foreign Application Priority Data

JP	2021-057931	Mar. 30, 2021
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Related U.S. Application Data

continuation parent-doc WO PCT/JP2022/000586 20220111 PENDING child-doc US 18468034

Publication Classification

Int. Cl.: B62D1/04 (20060101); G06F3/01 (20060101); G06F3/0362 (20130101); G06F3/044 (20060101)

U.S. Cl.:

CPC B62D1/046 (20130101); G06F3/016 (20130101); G06F3/0362 (20130101); G06F3/044 (20130101);

Field of Classification Search

CPC: B62D (1/046); G06F (3/016); G06F (3/0362); G06F (3/044)

USPC: 701/36

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2011/0220480	12/2010	Yorino	N/A	N/A
2015/0253950	12/2014	Onaka	345/184	G06F 3/0482
2019/0333720	12/2018	Ka	N/A	G06F 3/0362

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2011-187378	12/2010	JP	N/A
2015-170119	12/2014	JP	N/A
2015-205678	12/2014	JP	N/A
2019-160323	12/2018	JP	N/A

OTHER PUBLICATIONS

International Search Report for PCT/JP2022/000586 mailed on Mar. 29, 2022. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Application No. PCT/JP2022/000586, filed on Jan. 11, 2022, and designating the U.S., which is based upon and claims priority to Japanese Patent Application No. 2021-057931, filed on Mar. 30, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND

Technical Field

(1) The present invention relates to an input device.

Description of Related Art

(2) Hitherto, there has been a vehicle-mountable input device including: a housing including a supporting portion; a first operation knob including a supported portion swingably supported by the supporting portion; and a switch to be operated in response to a swinging operation on the first operation knob, wherein the first operation knob includes, in one lateral side thereof intersecting with a swinging shaft line, a cutout portion having an opening portion intersecting with the swinging shaft line, and a rotating knob generally referred to as a thumb wheel is situated in the cutout portion as a second operation knob corresponding to another switch. The rotational center shaft of the rotating knob is set to intersect with the one lateral side of the first operation knob (for example, see Japanese Laid-Open Patent Publication No. 2011-187378).

SUMMARY

(3) Here, in existing vehicle-mountable input devices, the operation that can be input via the rotating knob is limited to one type. Because a rotatable input unit such as a rotating knob has a good operability, it becomes even more handy when it can allow a plurality of types operations.

(4) Hence, an object is to provide an input device including a rotatable input unit by which it is possible to perform a plurality of types of operations.

(5) An input device according to an embodiment of the present invention includes: an input unit that is rotatable about a rotational center shaft; an electrostatic sensor situated around the input unit; and an operation position detecting unit configured to detect, based on an output of the electrostatic sensor, a position of an operational input that is performed on the input unit or around the input unit.

(6) It is possible to provide an input device including a rotatable input unit by which it is possible to perform a plurality of types of operations.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1A** to **1C** are views illustrating a steering wheel **10** mounted with an input device **100** according to an embodiment;

(2) FIG. **2** is a view illustrating an example of a configuration of the input device **100**;

(3) FIG. **3** is a block diagram illustrating the configuration of the input device **100**;

(4) FIG. **4** is a flowchart illustrating an example of a process performed by a control device **160**; and

(5) FIG. **5** is a flowchart illustrating another example of a process performed by the control device **160**.

DETAILED DESCRIPTION

(6) An embodiment to which an input device of the present invention is applied will be described below.

Embodiment

(7) FIGS. **1A** to **1C** are views illustrating a steering wheel **10** mounted with an input device **100** according to an embodiment. As illustrated in FIG. **1A**, the steering wheel **10** is mounted on, for example, a vehicle, and the input device **100** is mounted inside a spoke **11**. A rotating knob **110** of the input device **100** is exposed through an opening portion **11A** in the surface of the spoke **11**. The rotating knob **110** is an example of the input unit. For example, a driver of the vehicle can perform an operation of rotating the rotating knob **110** by the thumb of his/her left hand as indicated by a double-headed arrow in FIG. **1B** while gripping the steering wheel **10** by the left hand. The driver is an example of a user of the input device **100**. The rotating knob **110** can be operated by any other fingers of the left hand than the thumb. However, an embodiment in which the rotating knob **110** is operated by the thumb of the left hand will be described here.

(8) As illustrated in FIG. 1C, the input device **100** includes an electrostatic sensor **120** provided around the rotating knob **110**. In FIG. 1C, for example, five electrostatic sensors **120** are arranged horizontally, both above and below the rotating knob **110**, respectively. The electrostatic sensors **120** are configured to detect which position around the rotating knob **110** the thumb is touching.

(9) FIG. 2 is a view illustrating an example of the configuration of the input device **100**. The input device **100** includes a housing **101**, a rotating knob **110**, electrostatic sensors **120**, wiring boards **130**, an encoder **140**, and an actuator **150**. The encoder **140** is an example of a rotation position detecting unit, and the actuator **150** is an example of a tactile impression generating unit configured to generate a tactile impression.

(10) Here, an XYZ coordinates system is defined for description. A direction (X direction) parallel with an X axis, a direction (Y direction) parallel with a Y axis, and a direction (Z direction) parallel with a Z axis are orthogonal to one another. Observation in a plan-view perspective represents observation on the XY plane. The foregoing FIGS. 1A to 1C is illustrated by observation in the plan-view perspective (on the XY plane) in this XYZ coordinates system.

(11) The housing **101** is, for example, a box-shaped case made of a resin and having a rectangular parallelepiped shape. The longer direction of the housing **101** is in the X direction, and the shorter direction thereof is in the Y direction. The housing **101** has an opening portion **101A**, which is open along the entire length of the housing **101** in the +Z direction. In addition to the rotating knob **110**, the electrostatic sensors **120**, the wiring boards **130**, the encoder **140**, and the actuator **150**, a non-illustrated control device is provided on the housing **101**.

(12) The rotating knob **110** is provided in the housing **101**, and includes a rotating shaft **111** that is placed along the XY plane. The rotating shaft **111** is an example of a rotational center shaft of the input unit. The direction (X direction) in which the rotating shaft **111** extends is the same as the axial direction of the rotating knob **110**. The rotating shaft **111** is rotatably held on the housing **101**, and extends in the X direction. A rotating shaft of the encoder **140** is secured to the +X direction side of the rotating shaft **111**. The rotating shaft of the encoder **140** is secured to a driving shaft **151** of the actuator **150**. An edge portion of the rotating knob **110** at the +Z direction side projects from the housing **101** to the +Z direction side. The rotating knob **110** having such a configuration constitutes an input mechanism generally referred to as a thumb wheel. Since the rotating shaft **111** is formed along the plane (XY plane) in which the surface of the housing **101** exists, and since the edge portion of the rotating knob **110** at the +Z direction side projects from the surface of the housing **101** to the +Z direction side, an operator can perform an operation of rotating the rotating knob **110** relative to the surface of the housing **101** by stroking the surface of the housing **101** along one direction.

(13) The electrostatic sensors **120** are provided on the surface of each of two wiring boards **130**, such that five sensors are arranged along the X direction at both the +Y direction side and the -Y direction side of the rotating knob **110**, respectively. The +Y direction side of the rotating knob **110** is an example of one side of the one side and the other side across the rotating knob **110**, whereas the -Y direction side of the rotating knob **110** is an example of the other side of the one side and the other side across the rotating knob **110**.

(14) The electrostatic sensors **120** are, for example, metal electrodes having a square shape in the plan-view perspective. The electrostatic sensors **120** are situated on the back surface side of a cover of the spoke **11** of the steering wheel **10**, and are provided to detect the approaching of a thumb to the rotating knob **110**. More specifically, they are intended to detect a position of a thumb's operation with respect to the rotating knob **110**, i.e., which position of or around the rotating knob **110** is operated by the thumb. Because it is only necessary that the electrostatic sensors **120** be provided anywhere on the +X direction side of the rotating shaft **111** and anywhere on the -X direction side of the rotating shaft **111** around the rotating knob **110** in the plan-view perspective, the electrostatic sensors **120** may be provided at the exactly lateral sides of the rotating knob **110** at the +X direction side and the -X direction side. Here, an embodiment in which the electrostatic

sensors **120** are provided at the +Y direction side and the -Y direction side of the rotating knob **110** will be described. However, the electrostatic sensors **120** may be provided at either one of the +Y direction side and the -Y direction side. Moreover, the electrostatic sensors **120** are exemplified by separate electrostatic sensors. However, for example, those provided at the +Y direction side may be an integrated sensor configured to indicate a thumb position on a coordinates system.

(15) The wiring boards **130** are provided at the +Y direction side and the -Y direction side of the rotating knob **110** to extend in the X direction on the side walls **101B** of the housing **101**. The electrostatic sensors **120** are provided on the +Z direction side surfaces of the wiring boards **130**. The electrostatic sensors **120** are coupled to a non-illustrated control device through the wires of the wiring boards **130**.

(16) The encoder **140** includes a rotating shaft connected to the rotating shaft **111** of the rotating knob **110**. The rotating shaft of the encoder **140** is configured to rotate along with the rotating shaft **111** of the rotating knob **110**. The encoder **140** is provided to detect an amount of rotation of the rotating knob **110**, and is coupled to the non-illustrated control device through, for example, a non-illustrated wire. The encoder **140** is configured to detect a rotation position of the rotating shaft **111**, and send rotation position data indicating the rotation position to the control device.

(17) The actuator **150** includes a driving shaft **151** connected to the rotating shaft of the encoder **140**. For example, the actuator **150** is an electric motor, and can output a driving force, which is in a reverse direction or in a forward direction with respect to the rotation direction, to the driving shaft **151**. Since the driving shaft **151** is coupled to the rotating shaft **111** of the rotating knob **110** through the rotating shaft of the encoder **140**, the actuator **150** applies torque to the rotating shaft **111** of the rotating knob **110**.

(18) A driving control on the actuator **150** is performed by the control device. A reverse rotational force generated by the actuator **150** while a driver is rotating the rotating knob **110** by his/her thumb can provide a tactile impression that the rotational resistance of the rotating knob **110** has increased. A forward rotational force generated by the actuator **150** can provide a tactile impression that the rotational resistance of the rotating knob **110** has decreased. Here, for example, it is assumed that the actuator **150** generates a reverse rotational force.

(19) FIG. 3 is a block diagram illustrating the configuration of the input device **100**. FIG. 3 illustrates a control device **160** in addition to the electrostatic sensors **120**, the encoder **140**, and the actuator **150** of the input device **100**. The control device **160** is realized by a microcomputer including, for example, a Central Processing Unit (CPU), a Random Access Memory (RAM), a Read Only Memory (ROM), an input/output interface, and an internal bus.

(20) The control device **160** is coupled to the electrostatic sensors **120**, the encoder **140**, and the actuator **150**, and is coupled to an Electronic Control Unit (ECU) **20** through a cable **21**. A display **30** and a vehicle-mounted device **40** are coupled to the ECU **20**. FIG. 3 illustrates one electrostatic sensor **120** for simplicity. However, for example, ten electrostatic sensors **120** are actually coupled to the control device **160**. However, the number of the electrostatic sensors **120** is not limited to 10. Moreover, FIG. 3 illustrates one ECU **20**. However, a plurality of ECUs **20** may be coupled to the control device **160**, and there may be a plurality of vehicle-mounted devices **40**. There may also be a plurality of displays **30**.

(21) The display **30** is situated on, for example, a meter panel, a dashboard, or a center console of a vehicle. The ECU **20** is configured to control a content displayed on the display **30** in accordance with rotation position data acquired from the encoder **140**. A Graphical User Interface (GUI) image via which at least any vehicle-mounted device **40** is operated is displayed on the display **30**.

(22) The vehicle-mounted device **40** is, for example, an air conditioner, an audio device, a phone, or a cruise control function-equipped device of a vehicle. The vehicle-mounted device **40** is controlled by the ECU **20** in accordance with an operational input into the input device **100**. The ECU **20** is configured to display a GUI image matching the vehicle-mounted device **40** selected via the input device **100** on the display **30**. The GUI image is, for example, an image representing

temperatures and air flow rates of an air conditioner, sound volumes of an audio device and pieces of music to be selected on the audio device, contact addresses of phones, or speed settings of a cruise control function-equipped device along the circumference of a disk-shaped wheel, or a circular columnar thumb wheel-shaped image mimicking the rotating knob **110** of the input device **100**. Temperatures and air flow rates of the air conditioner, sound volumes of the audio device and pieces of music to be selected on the audio device, contact addresses of phones, and speed settings of the cruise control function-equipped device are examples of control targets.

(23) For example, a wheel image for temperature setting selection of the air conditioner represents numerals indicating temperature settings along the circumference of the wheel. By rotating the wheel in response to rotation of the rotating knob **110**, it is possible to select a temperature setting. A selected temperature setting is indicated by, for example, a numeral positioned at the front position in the circumference of the wheel. When displaying a thumb wheel-shaped image, numerals indicating temperature settings may be displayed along the side surface of a circular columnar shape. In this case, a setting to be selected is controlled to be positioned at the front position. In the following description, a wheel encompasses the concept described above. However, the concept described above is a mere example.

(24) The input device **100** is an input device configured to detect the position of an operation on the rotating knob **110** based on the capacitance of the electrostatic sensors **120**, to enable operating different control targets depending on whether to perform a rotating operation by contacting (touching) the +X direction side from the axial direction (X direction) center of the rotating knob **110** or to perform a rotating operation by contacting (touching) the -X direction side from the axial direction (X direction) center of the rotating knob **110**. That is, a driver can select a control target depending on which of the +X direction side and the -X direction side of the rotating knob **110** the driver will touch. Depending on a position of the input device **100** to be touched, it is possible to select one control target, which is to become an operation target, from a plurality of control targets.

(25) The +X direction side from the axial direction (X direction) center of the rotating knob **110** is an example of a first portion, and is also an example of a first-side half. The -X direction side from the axial direction (X direction) center of the rotating knob **110** is an example of a second portion other than the first portion, and is also an example of a second-side half. A control target to be selected by performing a rotating operation by touching the +X direction side from the axial direction (X direction) center of the rotating knob **110** is an example of a first control target. A control target to be selected by performing a rotating operation by touching the -X direction side from the axial direction (X direction) center of the rotating knob **110** is an example of a second control target.

(26) In the input device **100**, the actuator **150** is configured to generate a first rotational force that is reverse to a direction in which the rotating knob **110** is rotated by an operation when the rotating operation is performed by contacting (touching) the +X direction side from the axial direction (X direction) center of the rotating knob **110**, and the actuator **150** is configured to generate a second rotational force reverse to the rotating direction of the rotating knob **110** when the rotating operation is performed by contacting (touching) the -X direction side from the axial direction (X direction) center of the rotating knob **110**. For example, the first rotational force is higher than the second rotational force. This variation in the tactile impression to be provided to the driver's thumb via the rotating knob **110** is for enabling the driver, who performs a rotating operation by contacting the +X direction side or the -X direction side from the axial direction (X direction) center of the rotating knob **110**, to recognize one control target, which is to become the operation target, through a tactile impression. Moreover, by generating a rotational force intermittently, it is possible to generate a clicking impression as desired, during a rotating operation on the rotating knob **110**. Conversely, the first rotational force may be lower than the second rotational force.

(27) The input device **100** can continuously detect the displacement of the position at which an operation with respect to the rotating knob **110** is performed based on the capacitance of the

electrostatic sensors **120**, in order to enable an operation of moving the tip of a finger along the X direction while touching the rotating knob **110** without rotating the rotating knob **110** (hereinafter, this operation is referred to as a stroking operation), and an operation of not moving the tip of a finger for a certain period of time while touching a surface on the boundary between the rotating knob **110** and the opening portion **11A** in the surface of the spoke **11** or the rotating knob **110** without rotating the rotating knob **110** (hereinafter, this operation is referred to as a long pressing operation). The certain period of time is, for example, two seconds.

(28) For example, an embodiment in which it is possible to select temperature setting of an air conditioner as an operation target in response to touching the +X direction side of the rotating knob **110**, and it is possible to select the air flow rate of the air conditioner as an operation target in response to touching the -X direction side of the rotating knob **110** will be described below.

Moreover, an embodiment in which it is possible to adjust the sound volume of the audio device in response to performing a stroking operation on the input device **100**, and it is possible to turn off the sound volume of the audio device (or mute the audio device) in response to performing a long pressing operation on the input device **100** will be described below.

(29) The control device **160** includes a main control unit **161**, an operation position detecting unit **162**, a drive control unit **163**, a control unit **164**, and a memory **165**. The main control unit **161**, the operation position detecting unit **162**, the drive control unit **163**, and the control unit **164** are the functional block representation of the functions of a program executed by the control device **160**. The memory **165** is a functional representation of a memory of the control device **160**.

(30) The main control unit **161** is a processing unit configured to govern the control device **160**, and perform processes other than processes performed by the operation position detecting unit **162**, the drive control unit **163**, and the control unit **164**.

(31) The operation position detecting unit **162** is configured to detect the position (touch position) of a thumb that is contacting (touching) the input device **100** based on the capacitance of the ten electrostatic sensors **120**. The touch position detected by the operation position detecting unit **162** is an example of a detection result. The electrostatic sensor **120** that is the closest to the thumb has the maximum capacitance. Hence, for example, it is safe to detect that the thumb is present at a position corresponding to any electrostatic sensor **120** that has a capacitance higher than or equal to a threshold. The operation position detecting unit **162** sends position data indicating the touch position to the drive control unit **163**.

(32) When detecting a position (touch position) of the thumb that is contacting (touching) the input device **100** based on the capacitance of the ten electrostatic sensors **120**, the operation position detecting unit **162** may preferentially detect a position farther from the grip of the steering wheel over a position closer to the grip. This is because when one is trying to operate the +X direction side of the rotating knob **110** by his/her thumb, it can be assumed that the electrostatic sensors **120** at the -X direction side may also react at the same time as the electrostatic sensors **120** at the +X direction side react. When the electrostatic sensors **120** at both the +X direction side and the -X direction side react at the same time like this, by prioritizing detecting that the thumb is at the +X direction side (a region farther from the grip) over detecting that the thumb is at the -X direction side (a region closer to the grip), it is possible to realize a control suited to the intention of the operator. Moreover, it is optional to preferentially detect an operational input that moves away from the grip of the steering wheel **10**, over an operational input that moves closer to the grip. Such a preferential detection may optionally be performed because it can be assumed that, after a touching operation is performed by moving a thumb in a direction to be away from the grip and the touching operation is completed, the thumb will return in a direction to be closer to the grip in order to firmly grip the steering wheel **10**. This is for improving the touching operation detecting accuracy by not detecting a movement of a thumb during returning in a direction to be closer to the grip in order to firmly grip the steering wheel **10**, as a touching operation.

(33) The drive control unit **163** is configured to drive the actuator **150** in accordance with position

data that is input from the operation position detecting unit **162** and rotation position data indicating a rotation position, which is input from the encoder **140**. Here, for example, the drive control unit **163** controls the actuator **150** to generate a reverse first rotational force when the position data indicates the +X direction side from the axial direction (X direction) center of the rotating knob **110**, and controls the actuator **150** to generate a reverse second rotational force when the position data indicates the -X direction side from the axial direction (X direction) center of the rotating knob **110**. For example, the first rotational force is higher than the second rotational force. The drive control unit **163** determines the rotation direction of the rotating knob **110** based on a change in the rotation position data that is input from the encoder **140**, and controls the actuator **150** to generate a reverse rotational force.

(34) The control unit **164** is configured to output a control signal including both a touch position detected by the operation position detecting unit **162** and rotation data indicating either or both of: a change in the rotation position; and an amount of rotation, which is or are detected by the encoder **140** (this control signal is an example of a first control signal). That is, when a touching operation is performed and the rotating knob **110** is rotated, the control unit **164** outputs a control signal including rotation data and a touch position to the ECU **20**, and the ECU **20** selects a vehicle-mounted device **40**, which is to become the operation target, from the plurality of vehicle-mounted devices **40**, and controls the selected vehicle-mounted device **40** based on the control signal. Alternatively, the control unit **164** may select a vehicle-mounted device **40**, which is to become the operation target, from the plurality of vehicle-mounted devices **40** based on rotation data and a touch position and output a control signal, based on which the selected vehicle-mounted device **40** is to be controlled, to the ECU **20**, such that the ECU **20** may only need to perform control based on the control signal input from the control unit **164**.

(35) The control unit **164** detects a stroking operation and a long pressing operation based on a touch position detected by the operation position detecting unit **162**. The control unit **164** detects a stroking operation and a long pressing operation when a touching operation is performed and the rotating knob **110** is not operated.

(36) The control unit **164** outputs a control signal based on a stroking operation (this control signal is an example of a second control signal) to a vehicle-mounted device **40** that is previously designed to be operable by a stroking operation. The control unit **164** outputs a control signal based on a long pressing operation (this control signal is an example of a third control signal) to a vehicle-mounted device **40** that is previously designed to be operable by a long pressing operation. For example, any vehicle-mounted device **40** of the plurality of vehicle-mounted devices **40** may be determined as a vehicle-mounted device **40** that is operable by a stroking operation and a long pressing operation. Here, for example, an embodiment in which a vehicle-mounted device **40** serving as the audio device is a vehicle-mounted device **40** that is operable by a stroking operation and a long pressing operation will be described. A specific content of control by the control unit **164** will be described below with reference to FIG. 4 and FIG. 5.

(37) The control unit **164** optionally need not output a control signal based on which a vehicle-mounted device **40** is controlled, when no touch position is detected by the operation position detecting unit **162**, even when a rotation of the rotating knob **110** is detected by the encoder **140**. This makes it possible to prevent unintentional erroneous operations due to contact of, for example, belongings, which was not preventable with existing thumb wheel-type input devices.

(38) The memory **165** is configured to store, for example, programs and data that are needed for the main control unit **161**, the operation position detecting unit **162**, and the drive control unit **163** to perform processes, and, for example, data that are generated through the processes in addition.

(39) FIG. 4 is a flowchart illustrating an example of a process performed by the control device **160**. When the main control unit **161** starts the process, the following process is performed. Here, a process in a case of the control unit **164** outputting a control signal selecting a vehicle-mounted device **40**, which is to become the operation target, from among the plurality of vehicle-mounted

devices **40** based on a touch position detected by the operation position detecting unit **162** and a change in a rotation position detected by the encoder **140**, and the ECU **20** controlling the vehicle-mounted device **40** selected by the control signal output by the control unit **164** in accordance with the control signal will be described.

(40) The drive control unit **163** acquires rotation position data from the encoder **140** (step S1).

(41) The drive control unit **163** determines whether or not the rotating knob **110** is rotating, based on the rotation position data acquired in time series (step S2). When the rotation position indicated by the rotation position data acquired in time series is changing, this means that the rotating knob **110** is rotating.

(42) When the drive control unit **163** determines that the rotating knob **110** is rotating (S2: YES), the operation position detecting unit **162** acquires the capacitance of the ten electrostatic sensors **120** (step S3A).

(43) The operation position detecting unit **162** determines at which of the +X direction side and the -X direction side of the rotating knob **110** a touch position on the input device **100** is (step S4).

(44) When the operation position detecting unit **162** determines that the touch position is at the +X direction side, the drive control unit **163** drives the actuator **150** to generate a reverse first rotational force, and the control unit **164** sends position data indicating that the touch position on the rotating knob **110** is at the +X direction side to the ECU **20** (step S5A). As a result, the actuator **150** rotates the driving shaft **151** by the reverse first rotational force, and the ECU **20** sets temperature settings of the air conditioner as the operation target and displays a wheel indicating the temperature settings on the display **30**.

(45) The control unit **164** sends rotation position data indicating a rotation position at which the rotating knob **110** is stopped from being rotated to the ECU **20** (step S6A). As a result, the ECU **20** determines that a temperature setting, which is positioned at the front position of the wheel, is selected, and controls the temperature of the air conditioner accordingly. Through the steps described above, one series of processes is completed (end).

(46) In the step S4, when the operation position detecting unit **162** determines that the touch position is at the -X direction side, the drive control unit **163** drives the actuator **150** to generate a reverse second rotational force, and the control unit **164** sends position data indicating that the touch position on the rotating knob **110** is at the -X direction side to the ECU **20** (step S5B). As a result, the actuator **150** rotates the driving shaft **151** by the reverse second rotational force, and the ECU **20** sets the air flow rate of the air conditioner as the operation target and displays a wheel indicating air flow rates on the display **30**.

(47) The control unit **164** sends rotation position data indicating a rotation position at which the rotating knob **110** is stopped from being rotated to the ECU **20** (step S6B). As a result, the ECU **20** determines that an air flow rate, which is positioned at the front position of the wheel, is selected, and controls the air flow rate of the air conditioner accordingly. Through the steps described above, one series of processes is completed (end).

(48) When the drive control unit **163** determines in the step S2 that the rotating knob **110** is not rotating (S2: NO), the operation position detecting unit **162** acquires the capacitance of the ten electrostatic sensors **120** (step S3B).

(49) The operation position detecting unit **162** determines whether touching is performed or not (step S7).

(50) When determining that touching is performed (S7: YES), the operation position detecting unit **162** determines whether or not the touch position is changing (step S8). This is for determining whether a stroking operation is performed or not when the rotating knob **110** is not operated.

(51) When the operation position detecting unit **162** determines that the touch position is changing (S8: YES), the control unit **164** sends data indicating a stroking operation to the ECU **20** (step S9). As a result, the ECU **20** sets the sound volume of the audio device as the operation target, and adjusts the sound volume of the audio device in accordance with the stroking operation. Through

the steps described above, one series of processes is completed (end).

(52) When determining in the step **S8** that the touch position is not changing (**S8**: NO), the operation position detecting unit **162** determines whether or not the touch position is not changing for a certain period of time (step **S10**). This is for determining whether a long pressing operation is performed or not.

(53) When the operation position detecting unit **162** determines that the touch position is not changing for the certain period of time (**S10**: NO), the control unit **164** sends data indicating a long pressing operation to the ECU **20** (step **S11**). As a result, the ECU **20** sets the sound volume of the audio device as the operation target, and mutes the audio device in accordance with the long pressing operation. Through the steps described above, one series of processes is completed (end).

(54) When the operation position detecting unit **162** determines in the step **S7** that touching is not performed (**S7**: NO) and when the operation position detecting unit **162** determines in the step **S10** that the touch position is changing (**S10**: YES), the flow is terminated (end).

(55) In the present example, when it is determined in the step **S2** that the rotating knob **110** is not rotating (**S2**: NO) and it is determined in the step **S7** that touching is performed, the flow goes to the step **S8** to determine whether or not the touch position is changing. However, should the touch position be changing even though the rotating knob **110** is rotating, the rotation of the rotating knob **110** may optionally be ignored, determining that an unintentional rotation of the rotating knob **110** along with the stroking operation is occurring. That is, when a rotating operation causing the rotating knob **110** to rotate is performed at the same time as a stroking operation, the control unit **164** may ignore the rotating operation and send data indicating the stroking operation to the ECU **20**.

(56) For example, when a rotating operation on the rotating knob **110** is performed from the +Y direction side to the -Y direction side, the electrostatic sensors **120** positioned at the +Y direction side are detected earlier (or values to be detected become high earlier by being detected), the electrostatic sensors **120** positioned at the -Y direction side are detected afterwards (or values to be detected become high afterwards by being detected), a rotating movement of the rotating knob **110** is not continuously detected, and the electrostatic sensors **120** at the -Y direction side are detecting high values continuously for a predetermined period of time, this means that after the rotating operation on the rotating knob **110** is performed from the +Y direction to the -Y direction, the thumb is continuously being positioned at the -Y direction side of the rotating knob **110**. When such an operation is detected, for example, the control unit **164** may output a control signal instructing the rotating knob **110** to perform a rotating movement repeatedly many times (instructing the rotating knob **110** to continuously rotate) (this control signal is an example of a fourth control signal). The +Y direction side of the rotating knob **110** is an example of one side of one and the other sides across the rotating knob **110**, and the -Y direction side of the rotating knob **110** is an example of the other side of one and the other sides across the rotating knob **110**. For example, when a rotating operation on the rotating knob **110** is performed from the +Y direction side to the -Y direction side, a rotating movement of the rotating knob **110** is not continuously detected afterwards, and the electrostatic sensors **120** at the -Y direction side are detecting high values continuously for a predetermined period of time, this likewise means that after the rotating operation on the rotating knob **110** is performed from the +Y direction to the -Y direction, the thumb is continuously being positioned at the -Y direction side of the rotating knob **110**. Also in this case, for example, the control unit **164** may output a control signal instructing the rotating knob **110** to perform a rotating movement repeatedly many times (instructing the rotating knob **110** to continuously rotate) (this control signal is an example of a fourth control signal). Moreover, when such an operation is performed with the +Y direction side and the -Y direction side interchanged, the control unit **164** may output a control signal instructing the rotating knob **110** to perform a rotating movement repeatedly many times in a direction reverse to the direction in the above-described case (instructing the rotating knob **110** to continuously rotate) (this control signal is an

example of a fourth control signal).

(57) The control device **160** may perform the process illustrated in FIG. 5 instead of the process illustrated in FIG. 4. FIG. 5 is a flowchart illustrating another example of the process performed by the control device **160**. In the process illustrated in FIG. 4, detection of a rotation is performed first, and detection of a touching operation is performed subsequently. In FIG. 5, detection of a touching operation is performed first, and detection of a rotation is performed subsequently. When the main control unit **161** starts the process, the following process is performed. In the process of FIG. 5, a long pressing operation at the +X direction side is an operation of muting the audio device, and a long pressing operation at the -X direction side is an operation of returning the muted sound volume of the audio device to the sound volume before muting.

(58) The operation position detecting unit **162** acquires the capacitance of the ten electrostatic sensors **120** (step S21).

(59) The operation position detecting unit **162** determines whether and at which of the +X direction side and the -X direction side of the rotating knob **110** a touching operation on the input device **100** is performed (step S22).

(60) When determining in the step S22 that a touching operation is performed and at the +X direction side, the operation position detecting unit **162** determines whether and at which of the +X direction side and the -X direction side of the rotating knob **110** it was determined in the step S22 performed one control cycle before that a touching operation was performed on the input device **100** (step S23A).

(61) When the operation position detecting unit **162** determines that a touching operation was performed and at the +X direction side of the rotating knob **110**, the control unit **164** acquires rotation position data from the encoder **140** (step S24A).

(62) The control unit **164** determines whether or not a rotation position is changing, based on the acquired rotation position data (step S25A). This is for determining whether or not a rotating operation is performed on the rotating knob **110**.

(63) When the control unit **164** determines that a rotation position is changing (S25A: YES), the drive control unit **163** controls the actuator **150** to generate a reverse first rotational force, and the control unit **164** sends position data indicating that the touch position on the rotating knob **110** is at the +X direction side to the ECU **20** (step S26A). As a result, the actuator **150** rotates the driving shaft **151** by the reverse first rotational force, and the ECU **20** sets temperature settings of the air conditioner as the operation target and displays a wheel indicating temperature settings on the display **30**.

(64) The control unit **164** sends rotation position data indicating a rotation position at which the rotating knob **110** is stopped from being rotated to the ECU **20** (step S27A). As a result, the ECU **20** determines that a temperature setting, which is positioned at the front position of the wheel, is selected, and controls the temperature of the air conditioner accordingly. Through the steps described above, one series of processes is completed (end).

(65) When the control unit **164** determines in the step S25A that a rotation position is not changing (S25A: NO), the operation position detecting unit **162** determines whether the touch operation is not changing for a certain period of time (step S28A). This is for determining whether a long pressing operation is performed or not.

(66) When the operation position detecting unit **162** determines that the touch position is not changing for a certain period of time (S28A: NO), the control unit **164** sends data indicating a long pressing operation at the +X direction side to the ECU **20** (step S29A). As a result, the ECU **20** sets the sound volume of the audio device as the operation target, and mutes the audio device in accordance with the long pressing operation. Through the steps described above, one series of processes is completed (end).

(67) In the step S28A, when the operation position detecting unit **162** determines that the touch position is changing (S28A: YES), one series of processes is completed (end).

(68) When it is determined in the step S23A that the operation position detecting unit **162** has determined in the step S22 performed one control cycle before that a touching operation was performed and at the $-X$ direction side of the rotating knob **110**, the control unit **164** sends data indicating a stroking operation to the ECU **20** (step S30A). As a result, the ECU **20** sets the sound volume of the audio device as the operation target, and adjusts the sound volume of the audio device in accordance with the stroking operation. For example, when a stroking operation is performed from the $-X$ direction side to the $+x$ direction side, the stroking operation is for turning up the sound volume. Through the steps described above, one series of processes is completed (end).

(69) When it is determined in the step S23A that the operation position detecting unit **162** has determined in the step S22 performed one control cycle before that there was no position at which touching was performed, one series of processes is completed (end).

(70) When determining in the step S22 that a touching operation is performed and at the $-X$ direction side, the operation position detecting unit **162** determines whether and at which of the $+X$ direction side and the $-X$ direction side of the rotating knob **110** it was determined in the step S22 performed one control cycle before that a touching operation was performed on the input device **100** (step S23B).

(71) When the operation position detecting unit **162** determines that a touching operation was performed and at the $-X$ direction side of the rotating knob **110**, the control unit **164** acquires rotation position data from the encoder **140** (step S24B).

(72) The control unit **164** determines whether or not the rotation position is changing based on the acquired rotation position data (step S25B). This is for determining whether or not a rotating operation is performed on the rotating knob **110**.

(73) When the control unit **164** determines that the rotation position is changing (S25B: YES), the drive control unit **163** drives the actuator **150** to generate a reverse second rotational force, and the control unit **164** sends position data indicating that the touch position on the rotating knob **110** is at the $-X$ direction side to the ECU **20** (step S26B). As a result, the actuator **150** rotates the driving shaft **151** by the reverse second rotational force, and the ECU **20** sets the air flow rate of the air conditioner as the operation target and displays a wheel indicating air flow rates on the display **30**.

(74) The control unit **164** sends rotation position data indicating a rotation position at which the rotating knob **110** is stopped from being rotated to the ECU **20** (step S27B). As a result, the ECU **20** determines that the air flow rate, which is positioned at the front position of the wheel, is selected, and controls the air flow rate of the air conditioner accordingly. Through the steps described above, one series of processes is completed (end).

(75) When the control unit **164** determines in the step S25B that the rotation position is not changing (S25B: NO), the operation position detecting unit **162** determines whether the touch position is not changing for a certain period of time (step S28B). This is for determining whether a long pressing operation is performed or not.

(76) When the operation position detecting unit **162** determines in the step S28B that the touch position is changing (S28B: YES), one series of processes is completed (end).

(77) When the operation position detecting unit **162** determines that the touch position is not changing for a certain period of time (S28B: NO), the control unit **164** sends data indicating a long pressing operation at the $-X$ direction side to the ECU **20** (step S29B). As a result, the ECU **20** sets the sound volume of the audio device as the operation target, and returns the sound volume of the audio device to the original sound volume from the mute setting. Through the steps described above, one series of processes is completed (end).

(78) When the operation position detecting unit **162** determines in the step S23B that it was determined in the step S22 performed one control cycle before that a touch position was at the $+X$ direction side of the rotating knob **110**, the control unit **164** sends data indicating a stroking operation to the ECU **20** (step S30B). As a result, the ECU **20** sets the sound volume of the audio

device as the operation target, and adjusts the sound volume of the audio device in accordance with the stroking operation. For example, when a stroking operation is performed from the +X direction side to the -X direction side, the stroking operation is for turning down the sound volume. Through the steps described above, one series of processes is completed (end).

(79) When the operation position detecting unit **162** determines in the step S22 that a touching operation is not performed (no touching), one series of processes is completed (end).

(80) When it is determined in the step S23B that the operation position detecting unit **162** has determined in the step S22 performed one control cycle before that there was no position at which touching was performed, one series of processes is completed (end).

(81) As described above, according to any of the processes of FIG. 4 and FIG. 5, a driver can select an operation target in accordance with which of the +X direction side and the -X direction side of the rotating knob **110** he/she touches when performing a rotating operation. For example, an operation of pressing the rotating knob **110** or an operation on a switch other than the rotating knob **110** are not necessary in order to select an operation target, and it is possible to select an operation target in accordance with which of the +X direction side and the -X direction side is touched when performing a rotating operation on the rotating knob **110**.

(82) Hence, it is possible to provide an input device **100** including a rotating knob **110** by which it is possible to perform a plurality of types of operations. Because it is possible to select an operation target in accordance with which of the +X direction side and the -X direction side is touched, the driver can drive with minimal attentiveness to the operations of the input device **100**. Hence, an input device **100** having an extremely good convenience of use can be provided. Moreover, because there is no need of performing, for example, a pressing operation on a selector switch provided separately in order to switch between operation targets, it is possible to reduce the number of times to operate the vehicle-mounted devices **40**, and the driver can feel an extremely good convenience of use when he/she is driving a vehicle.

(83) The control unit **164** selects a control target, which is to become the operation target, from a plurality of control targets based on a detection result of the operation position detecting unit **162**, and outputs a control signal based on which the control target, which is selected as the operation target based on the operational input, is to be controlled. Hence, it is possible to select a control target, which is to become the operation target, depending on the position at which a touching operation is performed, and an input device **100** having an extremely good convenience of use can be provided.

(84) Moreover, because the actuator **150** and the drive control unit **163** being provided such that the actuator **150** is driven to give different tactile impressions depending on the control target selected as the operation target, it is possible to recognize the control target through the tactile impression, and to operate a plurality of control targets by blind touching during driving.

(85) Moreover, because the actuator **150** applies torque to the rotating shaft **111** of the rotating knob **110**, it is possible to recognize the control target through the tactile impression felt when rotating the rotating knob **110**, and to easily select an operation target from a plurality of control targets.

(86) Moreover, because the control unit **164** selects different control targets as the operation target depending on whether a touch position is at the +X direction side from the axial direction (X direction) center of the rotating knob **110** or a touch position is at the -X direction side from the axial direction (X direction) center of the rotating knob **110**, it is possible to easily select a corresponding operation target depending on whether a touch position is at the +X direction side or at the -X direction side, and an input device **100** having an extremely good convenience of use can be provided.

(87) Moreover, because the control unit **164** selects different control targets as the operation target depending on whether a touching operation is performed and at a position that is at the +X direction-side half from the axial direction (X direction) center of the rotating knob **110** or a touching operation is performed and at a position that is at the -X direction-side half from the axial

direction (X direction) center of the rotating knob **110**, it is easy to select an operation target, and an input device **100** having an extremely good convenience of use can be provided.

(88) Because the encoder **140** is provided such that the control unit **164** can output a control signal including rotation data indicating an amount of rotation of the rotating knob **110** to a vehicle-mounted device **40**, it is possible to perform a desired operation on the vehicle-mounted device **40** depending on the amount of rotation of the rotating knob **110**, and an input device **100** having an extremely good convenience of use can be provided. Moreover, because the electrostatic sensors **120** are situated at either or both of the +Y direction side and the -Y direction side across the rotating shaft **111** of the rotating knob **110** in a plan-view perspective, it is possible to detect a touching operation performed on the rotating knob **110** and around the rotating knob **110** without fail.

(89) Moreover, because the electrostatic sensors **120** are situated along the rotating shaft **111** of the rotating knob **110** in a plan-view perspective, it is possible to detect without fail at which position in the extending direction of the rotating shaft **111** a touching operation is performed.

(90) Moreover, because the control unit **164** outputs a control signal, based on which any of the plurality of control targets is controlled, in accordance with a stroking operation performed along the extending direction of the rotating shaft **111** of the rotating knob **110** based on a detection result of the operation position detecting unit **162**, it is possible to control a vehicle-mounted device **40** also by performing a stroking operation along the rotating knob **110** without rotating the rotating knob **110**, and thus an input device **100** having an even better convenience of use can be provided.

(91) Moreover, because the control unit **164** outputs a control signal, based on which any of the plurality of control targets is controlled, in accordance with a long pressing operation performed with the touch position kept stationary without rotating the rotating knob **110** based on a detection result of the operation position detecting unit **162**, it is possible to control a vehicle-mounted device also by performing a long pressing operation on the rotating knob **110** or around the rotating knob **110** without rotating the rotating knob **110**, and an input device **100** having an even better convenience of use can be provided.

(92) Moreover, because the input device **100** is provided on the steering wheel **10** of a vehicle, a driver can operate a plurality of control targets by blind touching during driving, and an input device **100** satisfying both safety assurance during driving and operational convenience can be provided.

(93) The operation position detecting unit **162** detects an operational input that moves in a direction to be away from the grip of the steering wheel **10** preferentially over an operational input that moves in a direction to be closer to the grip. After a touching operation is performed by moving a thumb in a direction to be away from the grip and the touching operation is completed, the thumb returns in a direction to be closer to the grip in order to firmly grip the steering wheel **10**.

Therefore, by not detecting the movement of the thumb during returning as a touching operation, it is possible to improve the touching operation detecting accuracy.

(94) In the foregoing description, an embodiment, in which the input device **100** is attached on the steering wheel **10** such that the rotating shaft **111** of the rotating knob **110** is along the extending direction of the spoke **11** of the steering wheel **10**, has been described. However, the method of attaching the input device **100** on the steering wheel **10** is not limited to this attaching method. For example, the input device **100** may be attached such that the rotating shaft **111** is at an angle with respect to the extending direction of the spoke **11**, and the angle may be 90 degrees.

(95) In the foregoing description, an embodiment, in which the control target to be selected as the operation target at the +X direction-side half in the extending direction of the rotating shaft **111** of the input device **100** is different from that at the -X direction-side half, has been described. The extending direction of the rotating shaft **111** may be divided into three or more sections, to enable selecting of a control target, which is to become the operation target, from three or more control targets.

(96) In the foregoing description, an embodiment, in which the input device **100** is attached on the spoke **11** of the steering wheel **10**, has been described. The input device **100** may be attached at a location of the steering wheel **10** other than the spoke **11**, and may be attached at a part of a vehicle other than the steering wheel **10**. Moreover, the input device **100** may be attached on something other than a vehicle.

(97) The input device according to illustrative embodiments of the present invention has been described above. However, the present invention is not limited to the embodiments disclosed specifically, and various modifications and changes can be applied to the invention without departing the scope of the claims.

Claims

1. An input device, comprising: an input unit that is rotatable about a rotational center shaft; an electrostatic sensor situated around the input unit; a processor configured to: detect, based on an output of the electrostatic sensor, a position of an operational input that is performed on the input unit or around the input unit, and output a first control signal based on a detection result of the position of the operational input, wherein the input unit and the electrostatic sensor are situated on a spoke of a steering wheel of a vehicle, and when the detection result of the position of the operational input indicates that the operational input is detected at both of a region of the electrostatic sensor farther from a grip of the steering wheel and a region of the electrostatic sensor closer to the grip than is the region that is farther, the processor prioritizes the operational input detected at the region that is farther, and the processor controls a vehicle-mounted device that is connected to the input device to perform a predetermined function according to the first control signal based on the prioritized operational input.
2. The input device according to claim 1, further comprising: an actuator configured to generate a tactile impression that is provided to a user who performs the operational input on the input unit, wherein the processor is further configured to perform drive control on the actuator such that the tactile impression becomes different in accordance with the detection result of the processor.
3. The input device according to claim 2, wherein the actuator is configured to apply torque to the rotational center shaft.
4. The input device according to claim 2, wherein the processor performs drive control on the actuator such that the tactile impression when the detection result of the position of the operational input indicates a first part of the input unit in an extending direction of the rotational center shaft is different from the tactile impression when the detection result of the position of the operational input indicates a second part of the input unit in the extending direction of the rotational center shaft, the second part not being the first part.
5. The input device according to claim 4, wherein the first part is a first-side half of the input unit in the extending direction of the rotational center shaft, and the second part is a second-side half of the input unit in the extending direction of the rotational center shaft.
6. The input device according to claim 1, further comprising: an encoder configured to detect a rotation position of the input unit, wherein the processor outputs a control signal including rotation data indicating an amount of rotation of the input unit as the first control signal, the amount of rotation being derived from the rotation position.
7. The input device according to claim 1, wherein the processor outputs a second control signal matching the operational input performed along the rotational center shaft of the input unit, based on the detection result of the position of the operational input.
8. The input device according to claim 7, wherein the processor outputs a third control signal matching the operational input performed in a stationary state on the input unit or around the input unit, based on the detection result of the position of the operational input.
9. The input device according to claim 1, further comprising: an encoder configured to detect a

rotation position of the input unit, wherein the processor outputs a second control signal matching the operational input performed along the rotational center shaft of the input unit based on the detection result of the position of the operational input, and when outputting the second control signal in accordance with the operational input performed along the rotational center shaft of the input unit, the control unit disregards a control signal serving as the first control signal including rotation data indicating an amount of rotation of the input unit, the amount of rotation being derived from the rotation position.

10. The input device according to claim 1, wherein when the electrostatic sensor is seen in a plan-view perspective, the electrostatic sensor is situated at either or both of one side and the other side of the input unit across the rotational center shaft.

11. The input device according to claim 1, wherein when the electrostatic sensor is seen in a plan-view perspective, the electrostatic sensor is situated along the rotational center shaft of the input unit.

12. An input device, comprising: an input unit that is rotatable about a rotational center shaft; an electrostatic sensor situated around the input unit; and a processor configured to: detect, based on an output of the electrostatic sensor, a position of an operational input that is performed on the input unit or around the input unit, and output a first control signal based on a detection result of the position of the operational input, wherein when the electrostatic sensor is seen in a plan-view perspective, the electrostatic sensor is situated at either or both of one side and the other side of the input unit across the rotational center shaft, said input device further comprising: an encoder configured to detect a rotation position of the input unit, wherein the processor outputs a fourth control signal matching the operational input, based on the detection result of the position of the operational input, when outputting a control signal including rotation data indicating an amount of rotation of the input unit as the first control signal, the amount of rotation being derived from the rotation position, the processor outputs a control signal including rotation data indicating rotation to the one side as the fourth control signal, provided that the operational input performed in a stationary state at the one side of the input unit has been detected, and controls a vehicle-mounted device that is connected to the input device to perform a predetermined function according to the fourth control signal, and when outputting a control signal including rotation data indicating an amount of rotation of the input unit as the first control signal, the amount of rotation being derived from the rotation position, the processor outputs a control signal including rotation data indicating rotation to the other side as the fourth control signal, provided that the operational input performed in a stationary state at the other side of the input unit has been detected, and controls the vehicle-mounted device that is connected to the input device to perform another predetermined function according to the fourth control signal.

13. An input device, comprising: an input unit that is rotatable about a rotational center shaft; an electrostatic sensor situated around the input unit; and a processor configured to: detect, based on an output of the electrostatic sensor, a position of an operational input that is performed on the input unit or around the input unit, and output a first control signal based on a detection result of the position of the operational input wherein the input device is situated on a steering wheel of a vehicle, and wherein the processor preferentially detects the operational input that moves in a direction to be away from a grip of the steering wheel, over the operational input that moves in a direction to be closer to the grip and controls a vehicle-mounted device that is connected to the input device to perform a predetermined function according the first control signal based on the operational input that moves in the direction to be away from the grip of the steering wheel.
